Presenter notes

Slide 1

Timing: 0:00–3:00. Start with the employee's task, not the model name.

Branch employees need to find a specific operational requirement in long documents, often while a customer is waiting. The problem is not simply locating a keyword. The employee needs to understand the right action for the customer's particular situation, including qualifications and exceptions.

I built a working conversational assistant over all eleven supplied procedures. The collection contains roughly seventy-nine thousand extracted words and many tables. A business-account procedure is substantially larger than the others, so treating every document as one unit would make retrieval uneven.

The assistant generates an answer from retrieved passages and lets the employee inspect the evidence. It can carry a scenario into a follow-up, recognize a new topic, request clarification, and withhold a draft that fails checks. It does not perform transactions. Its knowledge boundary is the supplied snapshot; linked procedures and current production policy are outside that boundary.

The deliverables include the local application, explicit evaluation cases, raw response traces, architecture decisions, and this six-slide presentation with a timed demonstration. Implementation source is kept separate from the presentation materials, as the assignment requires.

Generative AI was used both to assist development and to generate runtime answers. The evaluation labels were AI-authored and source-checked, which means they are useful regression material but not independent ground truth. I will explain measured results and observed weaknesses rather than imply this is an approved production banking tool.

The leadership question is whether the approach is promising enough for independent operational validation. The objective for a future pilot would be faster access to correct guidance with clear evidence and appropriate escalation. No employee productivity savings have been measured yet. Transition: the next slide shows how the system creates that evidence boundary.

Slide 2

Timing: 3:00–6:30. Trace the complete path and explain the alternatives.

The offline path parses the eleven original HTML documents. The parser removes presentation-only material but preserves headings, instructions and table conditions. It produces a passage store with metadata and source hashes. BM25 term statistics and normalized embedding vectors are built from the same passages.

At question time, the application resolves whether the user is asking a new question or continuing an earlier scenario. It retrieves exact lexical matches and semantic matches, combines the rankings with reciprocal rank fusion, and uses a cross-encoder to rerank the candidates. Only a small selected evidence pack goes to the generative model.

The final generator is a locally executed Qwen3 four-billion-parameter model in four-bit GGUF form through llama.cpp. Quantization makes a larger model practical on the available CPU laptop. This was an empirical choice: the initial smaller model frequently added irrelevant details or produced drafts that were withheld. A stronger approved private or hosted model should be compared before any real deployment.

The output passes citation, number, length and claim-support screening. Those checks can fail in both directions. If checks fail, the interface explicitly shows source excerpts rather than disguising the rejected draft as a successful answer. If the retrieval signal is too weak, the system asks for clarification or abstains.

I chose RAG because the task is access to a small, revisable evidence collection. Fine-tuning does not itself give reliable citations or explicit deletion of obsolete procedures. Sending the entire corpus would be inefficient and would expose the generator to many unrelated conditions. Search alone is a valuable fallback but would not satisfy the requested generative chat experience.

The implementation uses a straightforward pipeline instead of an autonomous transaction agent. The corpus fits exact in-memory vector search, so a managed vector database is unnecessary for this exercise. Production entitlements, identity, signed source releases, concurrency and monitoring remain separate platform work. Transition: the next slide explains the most important content decision—keeping each instruction attached to its conditions.

Slide 3

Timing: 6:30–10:00. Use the monthly-testing example to make the data decisions tangible.

The input files are HTML exports with meaningful policy classes, not just standard headings. Parsing only h1 or h2 tags would miss much of the structure. The parser explicitly recognizes the title, publication date, major sections and subsections. It decodes UTF-8 directly to preserve punctuation and does not follow external links or execute scripts.

Tables contain many operational branches. The parser serializes rows with their column labels, carries conditions through rowspans, accounts for colspans, and includes nested table text once. Long rows repeat the leftmost condition in continuation passages. That reduces, but does not eliminate, the risk of an action being detached from its situation. A production solution should represent complex branching tables more formally and have subject-matter review of extraction semantics.

Narrative grouping targets roughly one hundred and ten whitespace words within a section, with twenty-word overlap for long blocks. These are word targets, not exact model-token counts. The generator also has a hard prompt-token budget. The initial larger passages diluted narrow rules; shorter passages improved retrieval on the regression questions.

The coin-machine procedure illustrates why distinctions matter: monthly testing is single control, while weekly balancing and random monthly cash counts are dual control. A broad answer about coin-machine tasks can be fluent and still omit the distinction the employee needs.

Keyword matching preserves exact transaction codes and identifiers. Dense search handles paraphrases. RRF combines ranking positions rather than comparing incomparable raw scores. The cross-encoder considers the question and passage jointly. On this small collection, exact vector search is fast and operationally simple.

Every passage has a source hash and stable identity. An identical rebuild preserves IDs; a changed source intentionally changes them. The text-preservation audit found all checked source paragraph five-word sequences, but that does not prove correct table semantics. Source versions and effective policy authority require more than a publication date. Transition: now show how these choices appear to an employee in conversation.

Slide 4

Timing: 10:00–13:00. The short dialogue summarizes source-checked facts; the demonstration will show actual generated responses and status.

The interface starts with three concrete example questions and a visible eleven-document collection. An employee can ask a question, inspect the returned sources, download an original procedure, ask a follow-up, switch topics or reset. Each source panel includes the section, publication text and passage identity. A trace and elapsed time are available without exposing implementation code.

Session memory is scoped to the browser conversation. Shared cached model objects do not contain user

histories. The resolver prioritizes explicit procedure IDs and recognizes a small set of named topics. A named new topic overrides previous scope. An elliptical follow-up can retain the prior question and procedure reference, but it always triggers new retrieval.

For the coin-machine example, the follow-up names a new task in the same procedure. The resolver retains the procedure scope but avoids repeating overnight-limit keywords that would distract retrieval from testing. For a notary question followed by 'what if the requester is physically present', the prior emailed-document scenario remains necessary context. These are different conversational cases and are tested separately.

The resolver is deterministic and transparent, not a claim of unrestricted conversational understanding. Ambiguous entities and long histories remain limitations. Short or pronominal questions without usable history prompt clarification. Unsupported topics clear the interface's prior topic context. A new-conversation control gives the user an explicit reset.

A helpful experience also includes honest uncertainty. The application distinguishes a generated answer, partial support, source excerpts after withholding a draft, and insufficient evidence. It does not display a made-up confidence percentage. Citation and NLI checks are screening signals, not guarantees. A source-only response can be useful, but it is counted as lost generated-answer coverage in evaluation.

The application uses hypothetical examples, blocks a few obvious identifier patterns and has no automatic persistent chat logging. Those choices are appropriate to this demo but are not comprehensive privacy controls. Transition: we should judge the experience using separate measures of retrieval, response quality and coverage.

Slide 5

Timing: 13:00–17:00. Do not describe any automatic proxy as independently established accuracy.

The included suite has thirty single-turn questions covering all eleven procedures, four two-turn conversations, and six guardrail questions. The labels were AI-authored from the supplied sources and evidence locators were checked against the extracted content. Nineteen questions are development cases and eleven are challenge cases, but after development inspection the challenge split is not a blind external holdout.

The retrieval comparison uses the same questions and specific relevant passages. Keyword search finds labelled evidence in 90.0% of top-five results, semantic search in 93.3%, and hybrid search with reranking in 100.0%. Reciprocal rank measures how early the first relevant passage appears. These figures do not establish that every applicable exception was retrieved, or that the generated answer is correct.

For generated responses, 73.3% of the thirty ordinary cases contain all required fact groups under the automatic proxy. That proxy can miss a correct paraphrase or pass an incorrect statement containing expected words. Source-only fallbacks count as failures of generated-answer coverage, even if their excerpts contain the answer. Across generated evaluations, 8 drafts were withheld. Guardrail expected-status agreement was 100.0%. Local serial CPU response latency was p50 6.7 seconds and p95 12.5 seconds, excluding initial model loading.

Early tests showed a notary response with a correct permission decision and an invented general exemption, and a metal-card answer that borrowed instructions from the wrong situation. This demonstrates why citations and word matching are not sufficient. NLI adds a useful screening signal but also rejects valid answers and can miss errors.

The next evaluation needs independent subject-matter reviewers scoring every material instruction for accuracy, relevance, completeness and conciseness. Critical errors—such as reversed prohibitions or wrong monetary thresholds—must be reported separately from averages. We also need realistic employee tasks, unseen scenarios, calibrated abstention, and latency under concurrency. Transition: these results justify further validation, not an unqualified operational rollout.

Slide 6

Timing: 17:00–20:00. End with a practical decision and concrete ownership.

The working prototype demonstrates end-to-end access to the supplied procedures, actual local generation, inspectable sources and bounded conversational context. It includes reproducible evaluation and recorded failures. It does not establish current policy authority, independent response correctness or enterprise readiness.

The first next step is a scoped validation effort with procedure owners and operational subject-matter experts. They should independently author and adjudicate scenarios, especially conditional tables, exceptions, eligibility and escalations. Review both generated answers and source-only fallbacks. A system that answers very little can appear safe while failing to help employees, so coverage must remain visible.

AI engineering should compare stronger approved model options on the same unseen scenarios and measure end-to-end latency and infrastructure cost. Retrieval improvements should focus on conditions, cross-references and exceptions, rather than adding an agent framework without a demonstrated need. Product and branch operations should compare task success and time to correct guidance against current search, with no claimed savings until measured.

Before any operational pilot, identity and security teams need SSO, document-level authorization before retrieval, session controls and approved data handling. Procedure owners need authoritative effective dates, supersession rules and rollback. Platform teams need concurrency controls, redacted audit logging, retention, monitoring and incident response. The business sponsor and governance owners define the release decision.

Proposed gates include zero critical unsupported instructions on an independently reviewed release suite, acceptable factual correctness and completeness, explicit refusal/fallback coverage, and an agreed answer-time target. Those are proposals for approval, not achieved enterprise service levels or guarantees of zero future errors.

Generative AI helped create this solution and also powers the assistant. The candidate must understand and be able to defend the design, its measured evidence and its limits. The presentation materials deliberately exclude source code, as requested. Transition to the fifteen-minute walkthrough: I will now show a factual question, a follow-up, an exception, a topic change, and how the application exposes or withholds evidence-supported answers.